HW5 Report

Group 26
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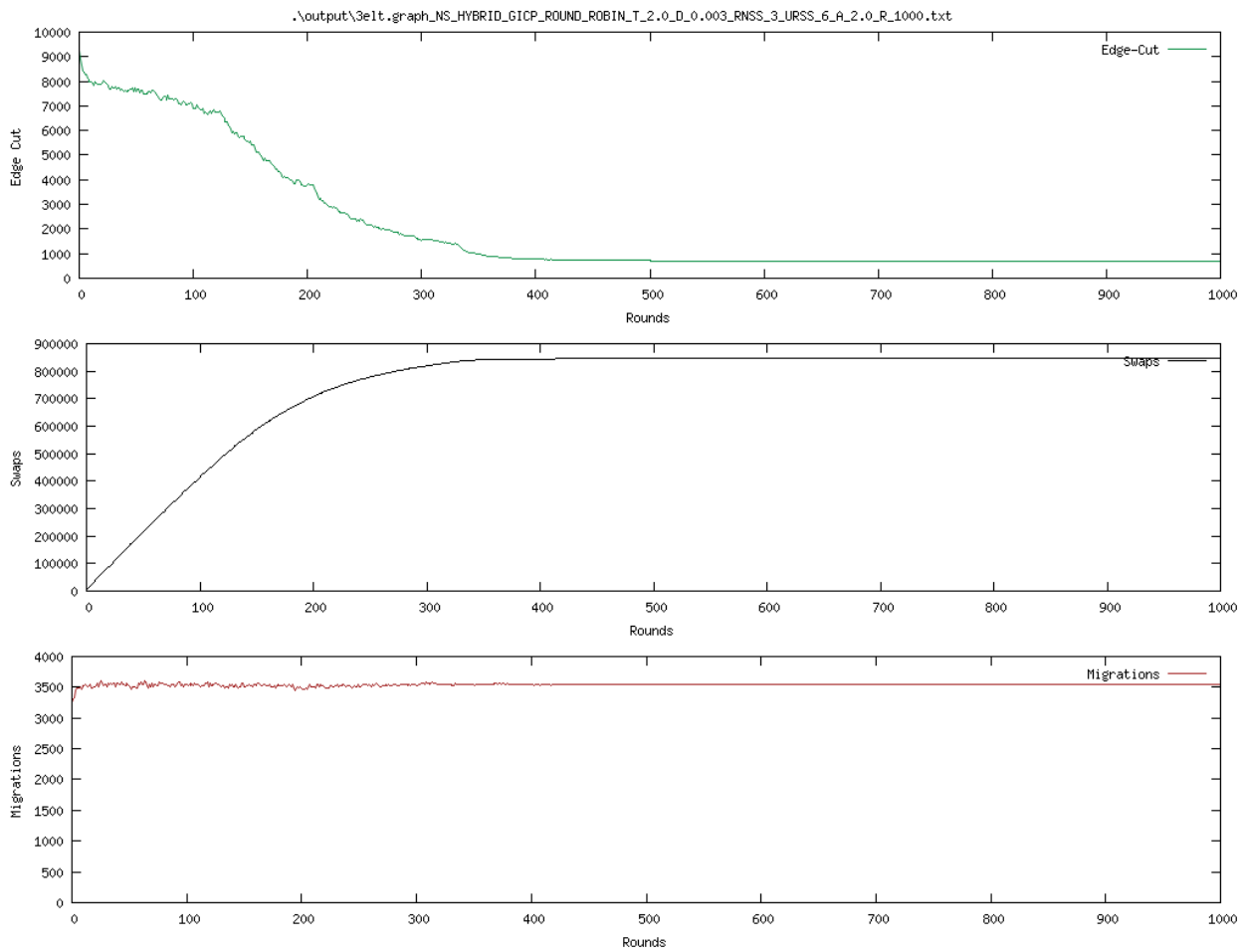
Task 1

We followed the scaffolding code for Ja-Be-Ja and implement the TODO part in sampleAnd Swap and findPartner method as is described in the JaBeJa paper.
Here we present our plot result for 3elt graph, add 20 and Facebook graph.

3elt

Graph Properties: 4,720 nodes, 13,722 edges
Runtime: 7 seconds (1000 rounds)
Convergence: Round 500
Final edge cut: 714 (5.2% of total edges)
Total swaps: 845,631
Migrations: 3,544 (75% of nodes)

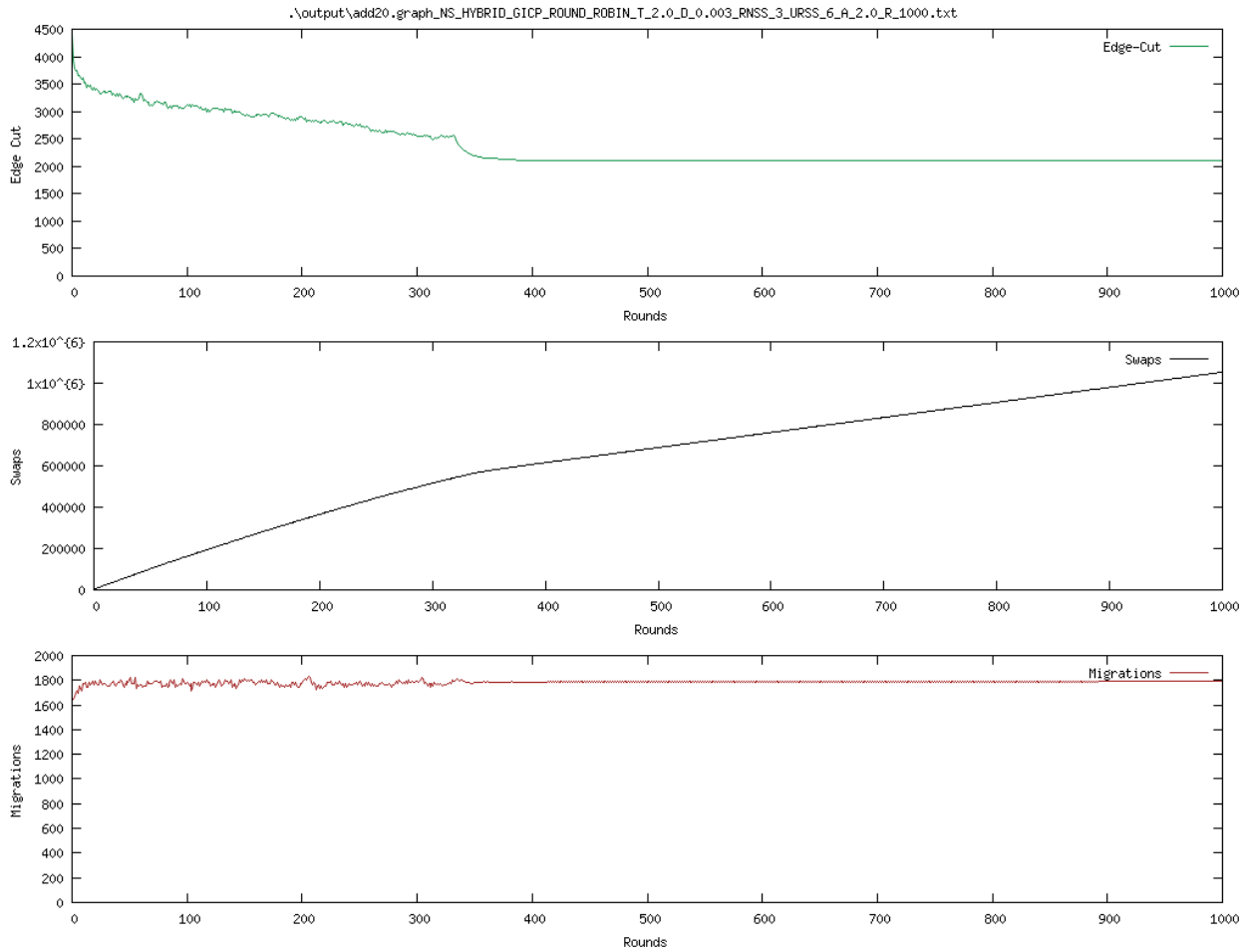
Analysis:
The 3elt graph demonstrates efficient partitioning behavior. The relatively low edge cut (5.2%) suggests the graph has a clear community structure that JaBeJa successfully exploited. The algorithm converged at round 500, which aligns with the temperature cooling schedule (T reaches 1.0 at round 333 with delta=0.003, T=2.0). After convergence, swaps stabilized as no further beneficial exchanges were found. The high migration rate (75%) indicates most nodes found better partitions during optimization, reflecting the algorithm's effectiveness on this well-structured graph.



Add20

Graph Properties: 2,395 nodes, 7,462 edges
Runtime: 5 seconds (1000 rounds)
Convergence: Round 409
Final edge cut: 2,102 (28.2% of total edges)
Total swaps: 1,052,366 (continuously growing)
Migrations: ~1,790 (continuously fluctuating)

The add20 graph exhibits problematic convergence behavior. The high edge cut ratio (28.2%) suggests weak community structure or imbalanced partitions. The continuous growth of swaps and fluctuating migrations indicate the algorithm is trapped in local optima oscillation. Even after temperature cooling, nodes keep swapping between equivalent partition configurations without improving the solution. This "churning" behavior occurs because: (1) the graph lacks clear boundaries between communities, (2) the acceptance criterion (newBenefit * T > old) still allows swaps when T=1 if benefits are equal, and (3) random sampling in hybrid policy creates non-deterministic partner selection, leading to repeated swaps between equally suboptimal states.

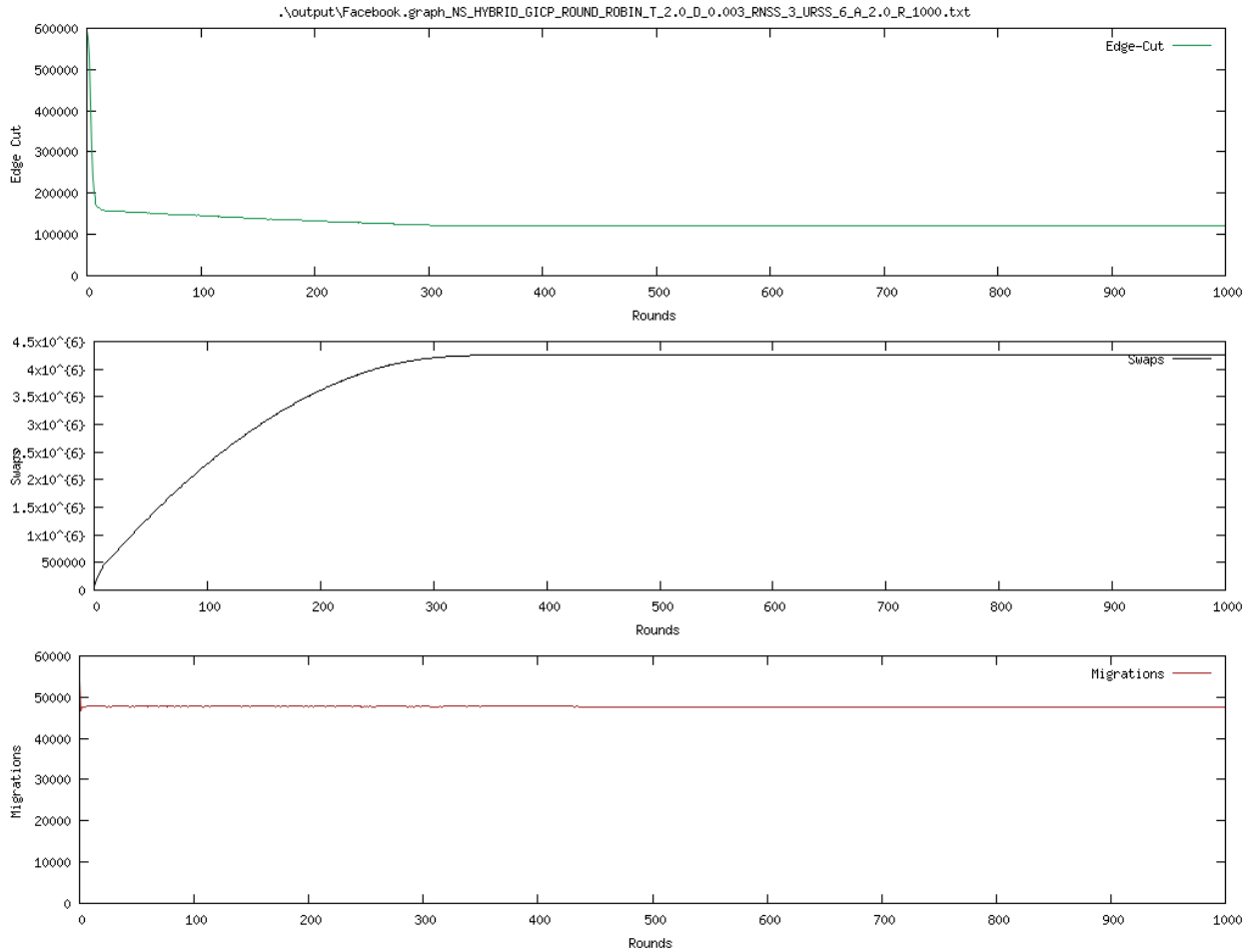


Facebook

Graph Properties: 63,731 nodes, 817,090 edges
Runtime: ~15 minutes (1000 rounds)
Convergence: Round 608
Edge cut trajectory: 593,189 → 157,139 (round 10) → 122,111 (round 300) → 120,427 (final)

The Facebook graph shows multi-phase convergence typical of large-scale social networks. The dramatic initial drop (73.5% reduction in 10 rounds) occurs during high-temperature exploration when the algorithm rapidly finds major community boundaries. The second phase (rounds 10-300) shows gradual refinement as temperature decreases, reducing edge cut by another 22%. The final phase (rounds 300-608) represents fine-tuning with minimal improvements (1.4%). This behavior reflects the hierarchical community structure of social networks: macro-communities are identified quickly, while micro-optimizations require many iterations. The later convergence (round 608 vs. temperature reaching 1.0 at round 333)

suggests the algorithm benefits from continued local optimization even without simulated annealing acceptance of worse solutions.



Task 2: Simulated Annealing

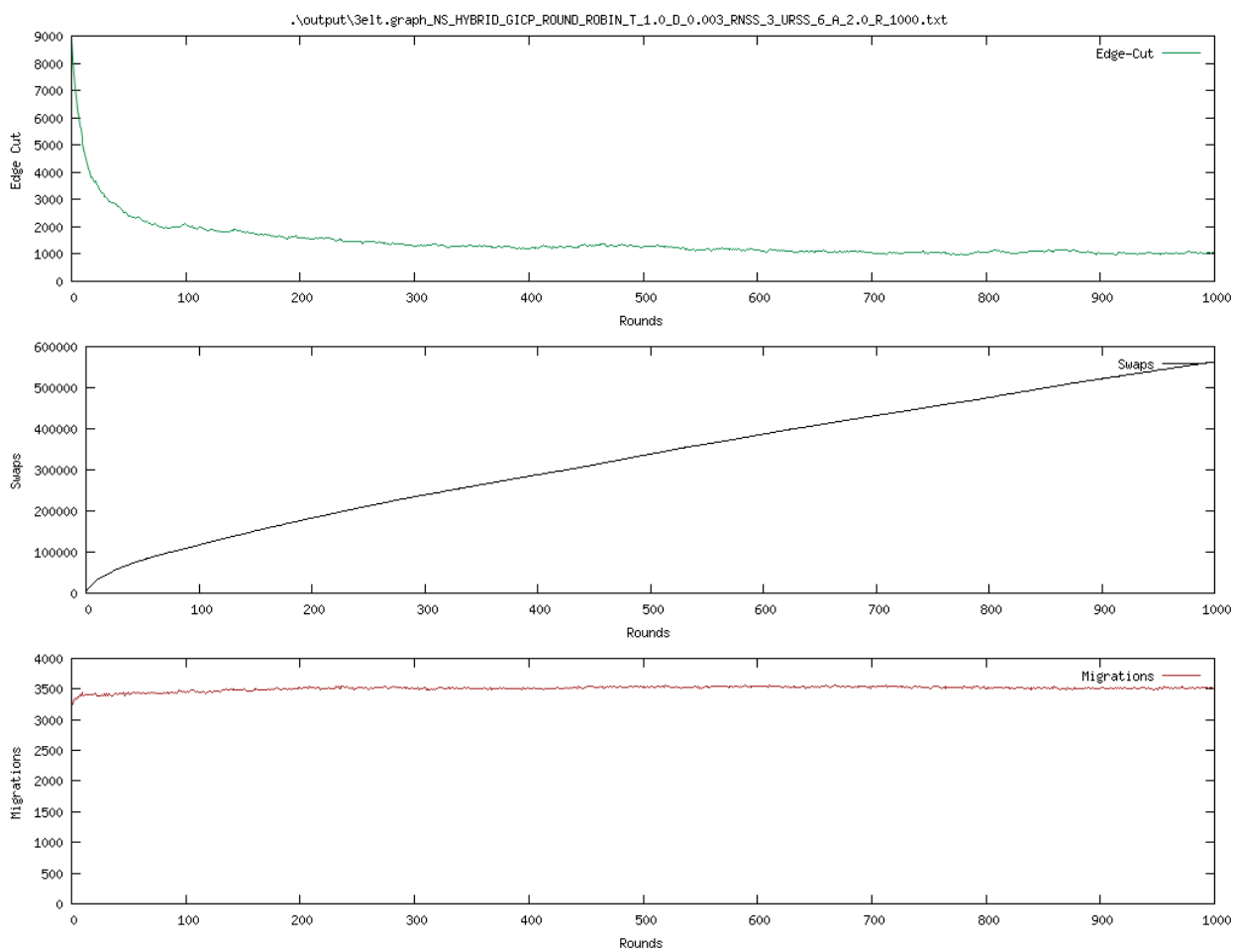
In task 2, we have implement the simulated annealing first and then tried the original jabeja algorithm followed by a simulated annealing.

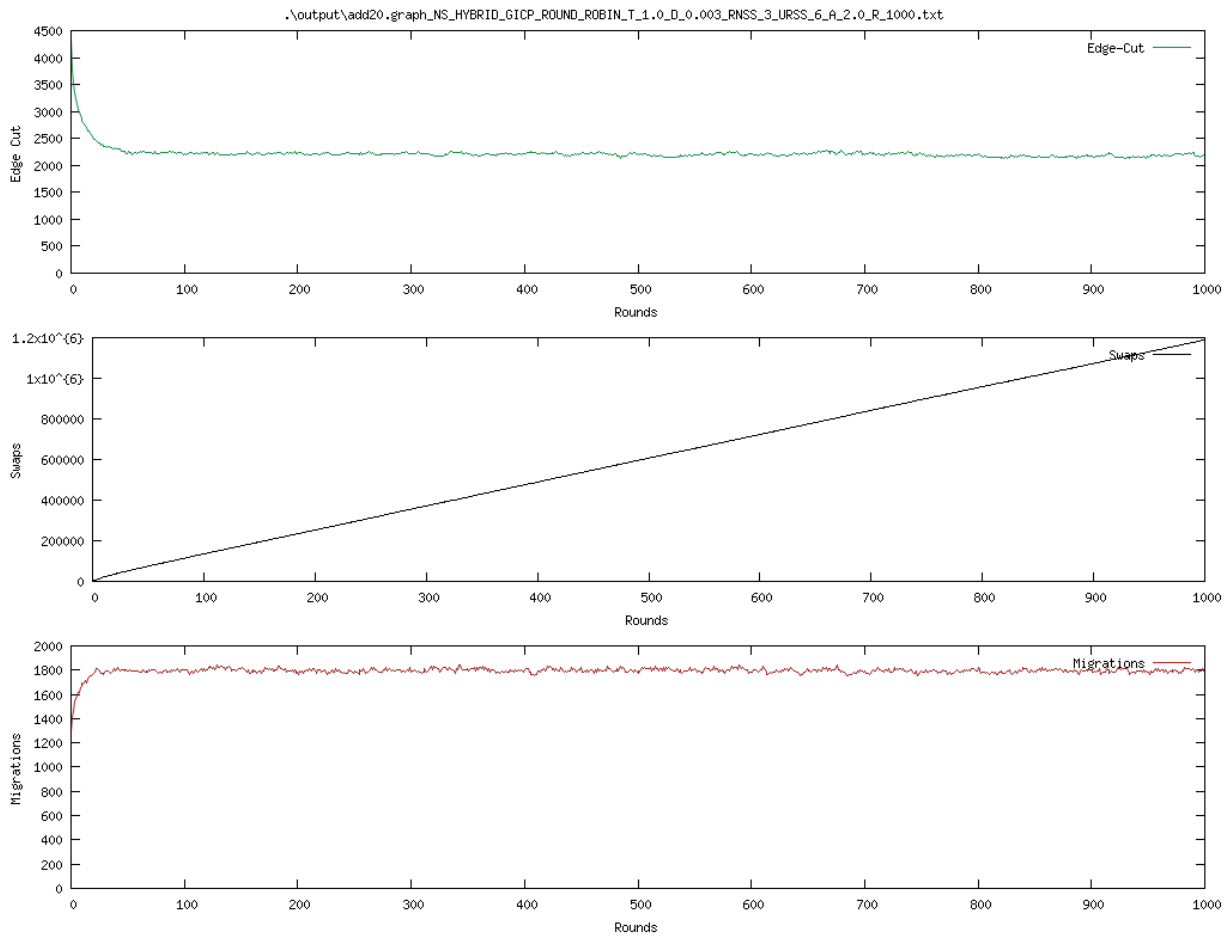
Only Simulated Annealing

In the first subsection of task 2, we modified the original jabeja: decide to exchange node color by acceptance probability function instead of temperature multiplying the cost. The main modification is the findPartner method.

```java
    if (newBenefit > old) {
      if (newBenefit > highestBenefit) {
        highestBenefit = newBenefit;
        bestPartner = nodeq;
      }
    } else {
      double ap = acceptanceProbability(old, newBenefit, this.T);
      double rand = Math.random();
      if (ap > rand) {
        if (newBenefit > highestBenefit) {
          highestBenefit = newBenefit;
          bestPartner = nodeq;
        }
      }
    }
  }
  return bestPartner;
```

We discovered the result is not as good as before in 3elt in terms of the final edge cut count. Although the swap-iteration relation suggests that the result is not converged eventually, as is shown in the edgecut-round, the edge cut drops faster than the original jabeja at the beginning.

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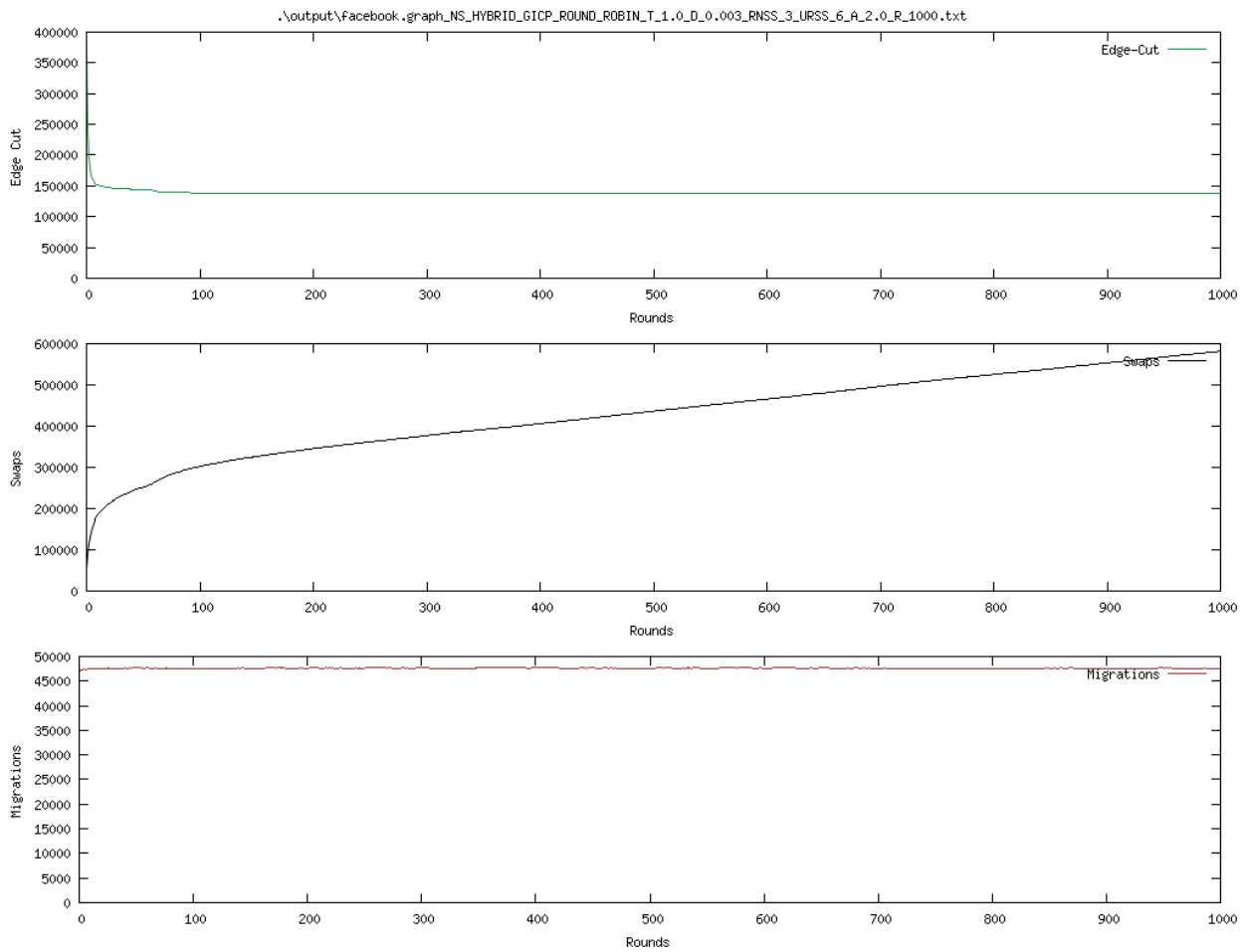
Edge-Cut
Edge Cut
Rounds
Swaps
Swaps
Rounds
Migrations
Migrations
Rounds

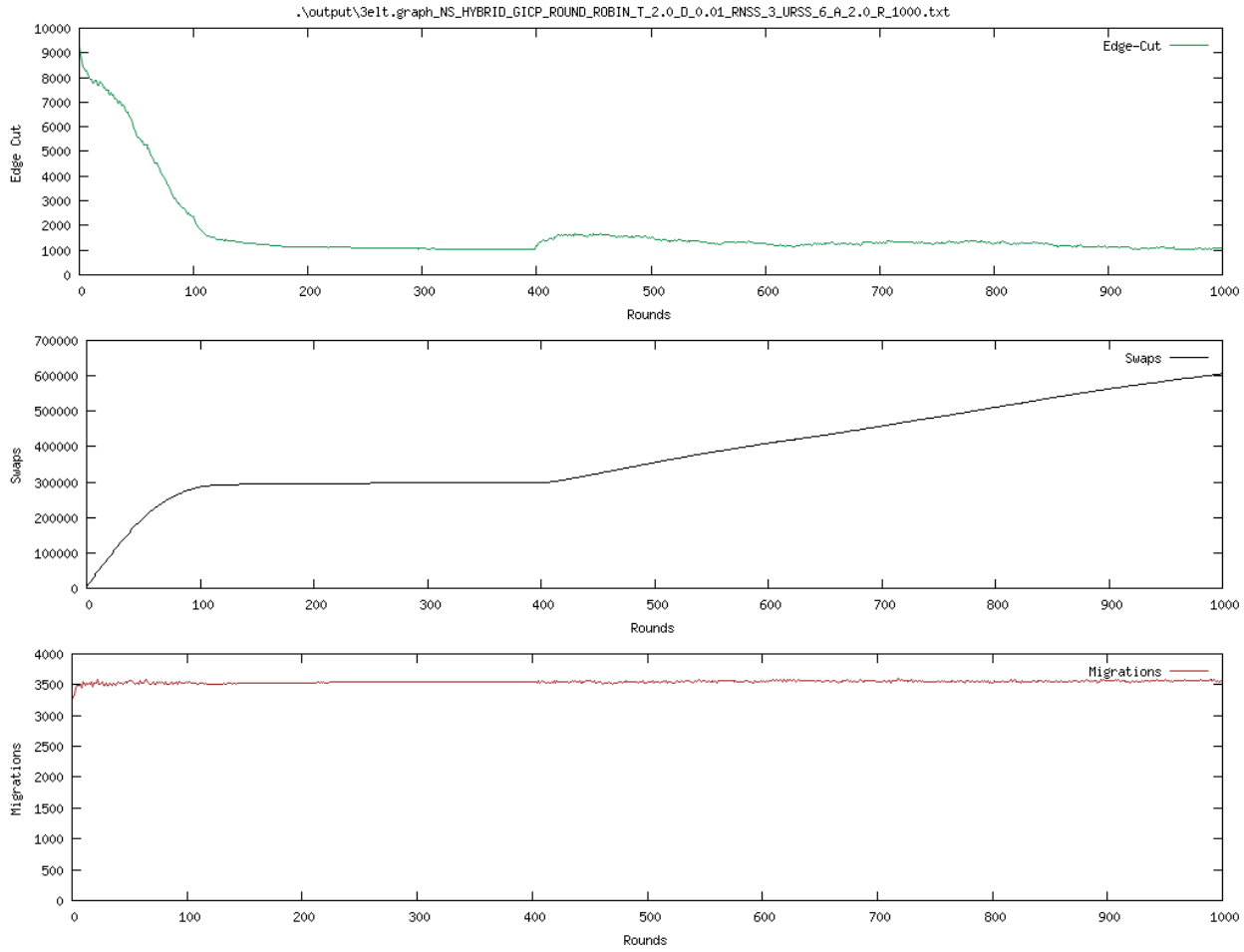


Original Ja-Be-Ja Followed by Simulated Annealing

In the second subsection, we restart the simulation annealing when the original JaBeJa has converges. It is based on the observation that when **T** reaches its final value (that is, no more bad swaps are allowed), Ja-Be-Ja converges to an edge cut rapidly, and the edge cut does not change over time. So as to step out of local optimum, the simulation annealing is introduced.

In the first 400 round, we followed Ja-Be-Ja and then we restart the simulated annealing to jump out of local optima.

After 400 round, the algorithm is reactivated, but the edge cut does not turn better.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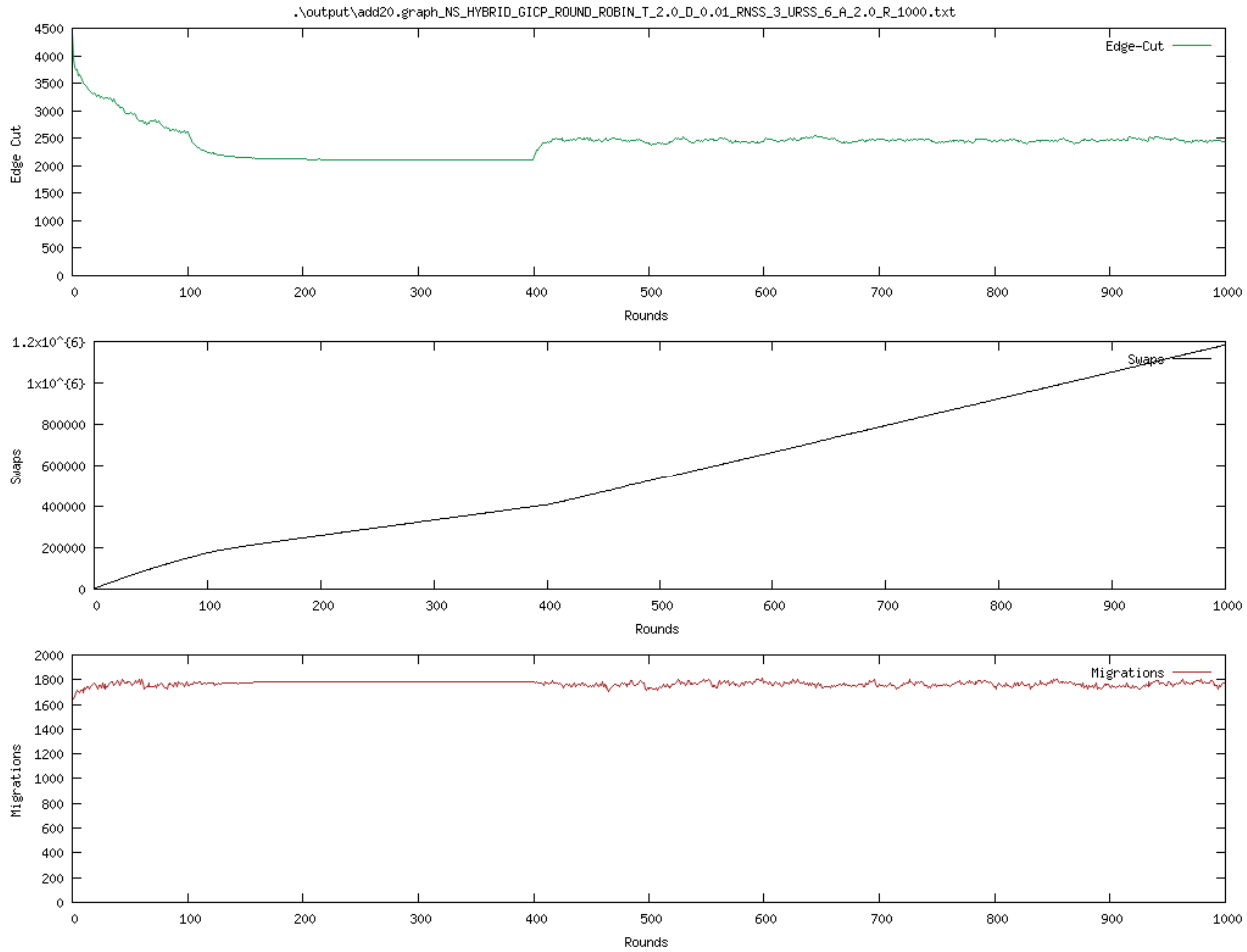
Edge-Cut
Edge Cut
Rounds
Swaps
Swaps
Rounds
Migrations
Migrations
Rounds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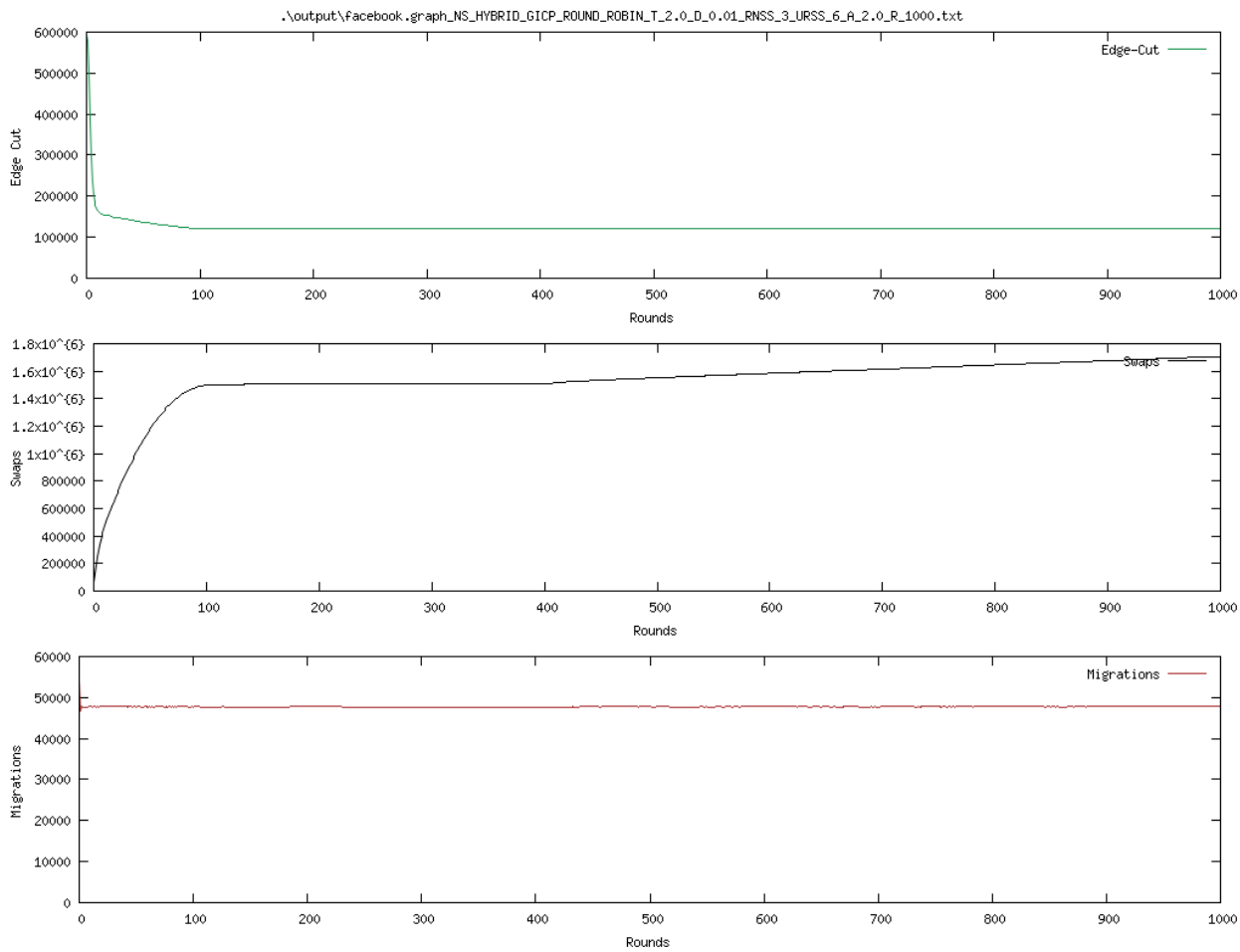
Edge-Cut
Edge Cut
Rounds
Swaps
Swaps
Rounds
Migrations
Migrations
Rounds

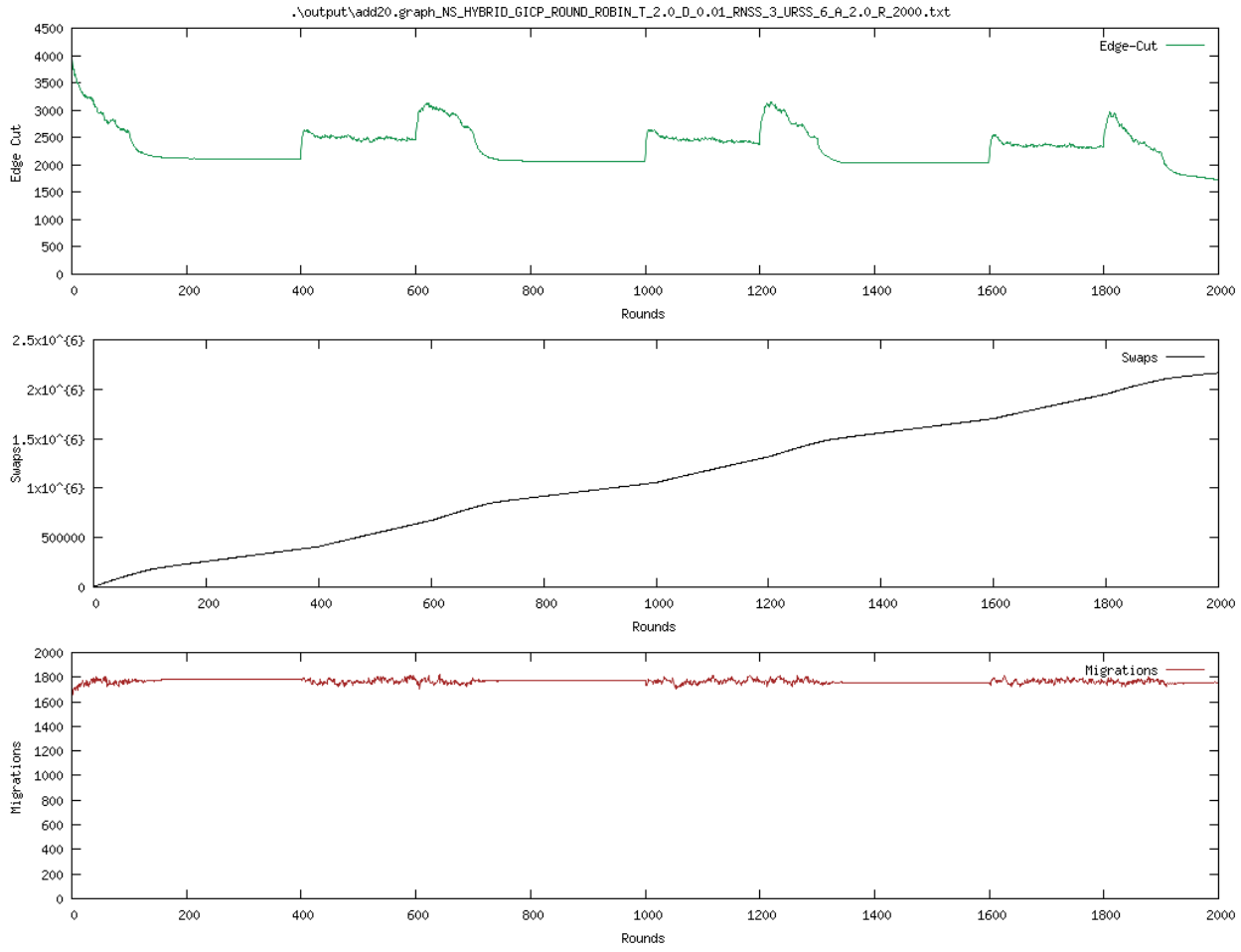


Bonus: Iterative Ja-Be-Ja & Simulated Annealing

Here we present our algorithm of multiple round of ja-be-ja&sa. In the second section of task 2, we observed that although simulated annealing with acceptance probability can successfully escape local optima, it sometimes omit the optimal situations as well. However, the benefit of the original Ja-Be-Ja is that it always converges to a optimal edge count. As a result, we can try multiple iterations between Ja-Be-Ja and simulated annealing to get closer to global optimal edge cut.

In the experiment, we set temperature as 2.0 as initial value, each round it reduce by 0.01, until it reaches a bottom of 1.0. The original Ja-Be-Ja runs 400 rounds and then it switches to simulated annealing for 200 rounds. The iteration goes on. We set 2000 rounds as an end to obtain the edge cut result.

The results show the validity of our algorithm. In add20 graph, the edge cut reduces from 1900 to 1700. The 3elt graph end it up in 900, but in the middle it reaches a low of 699, which suggest one further improvement will be adding a snapshot of the graph state to retrieve the optimal partition after the whole process is done.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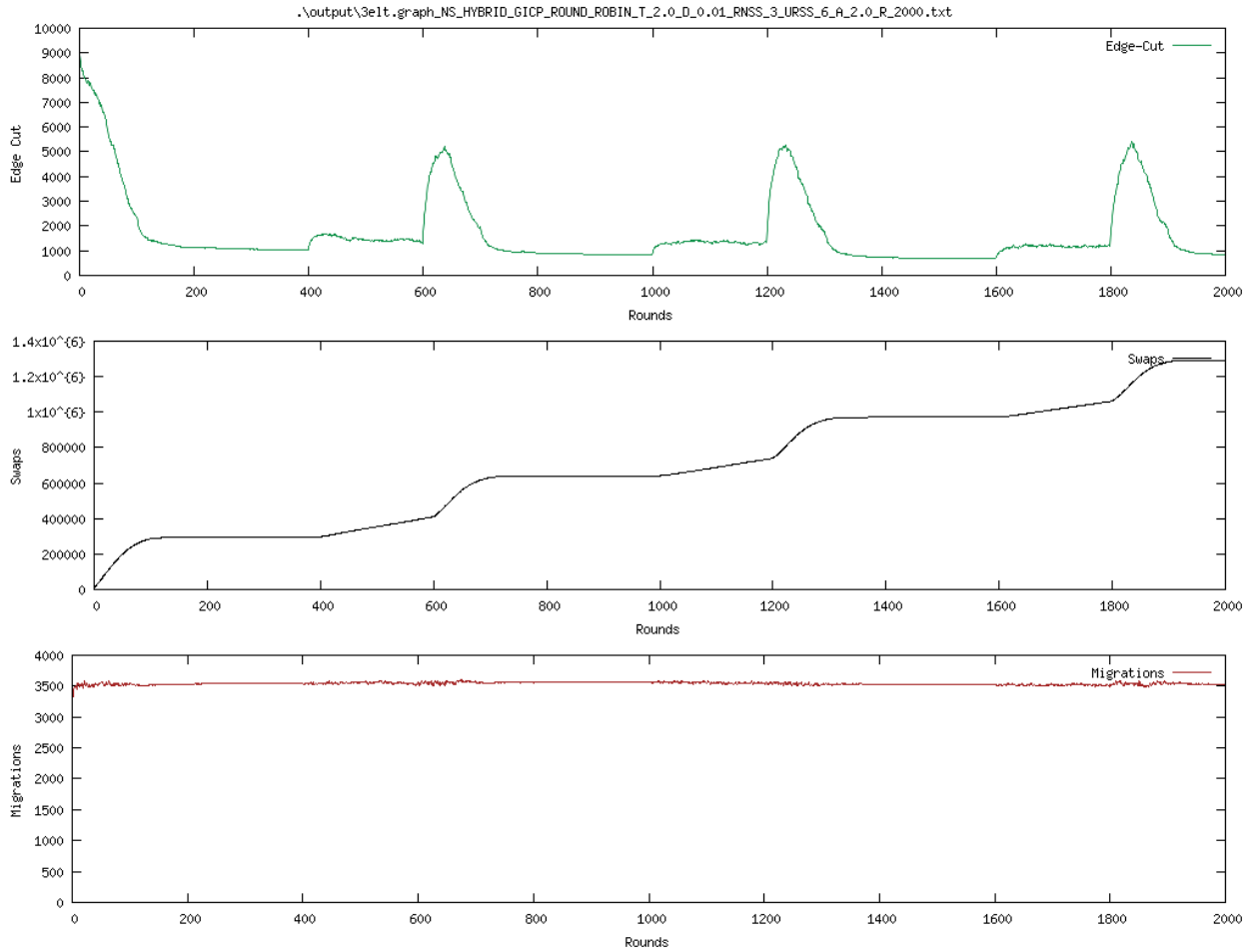
Edge-Cut
Edge Cut
Rounds
Swaps
Swaps
Rounds
Migrations
Migrations
Rounds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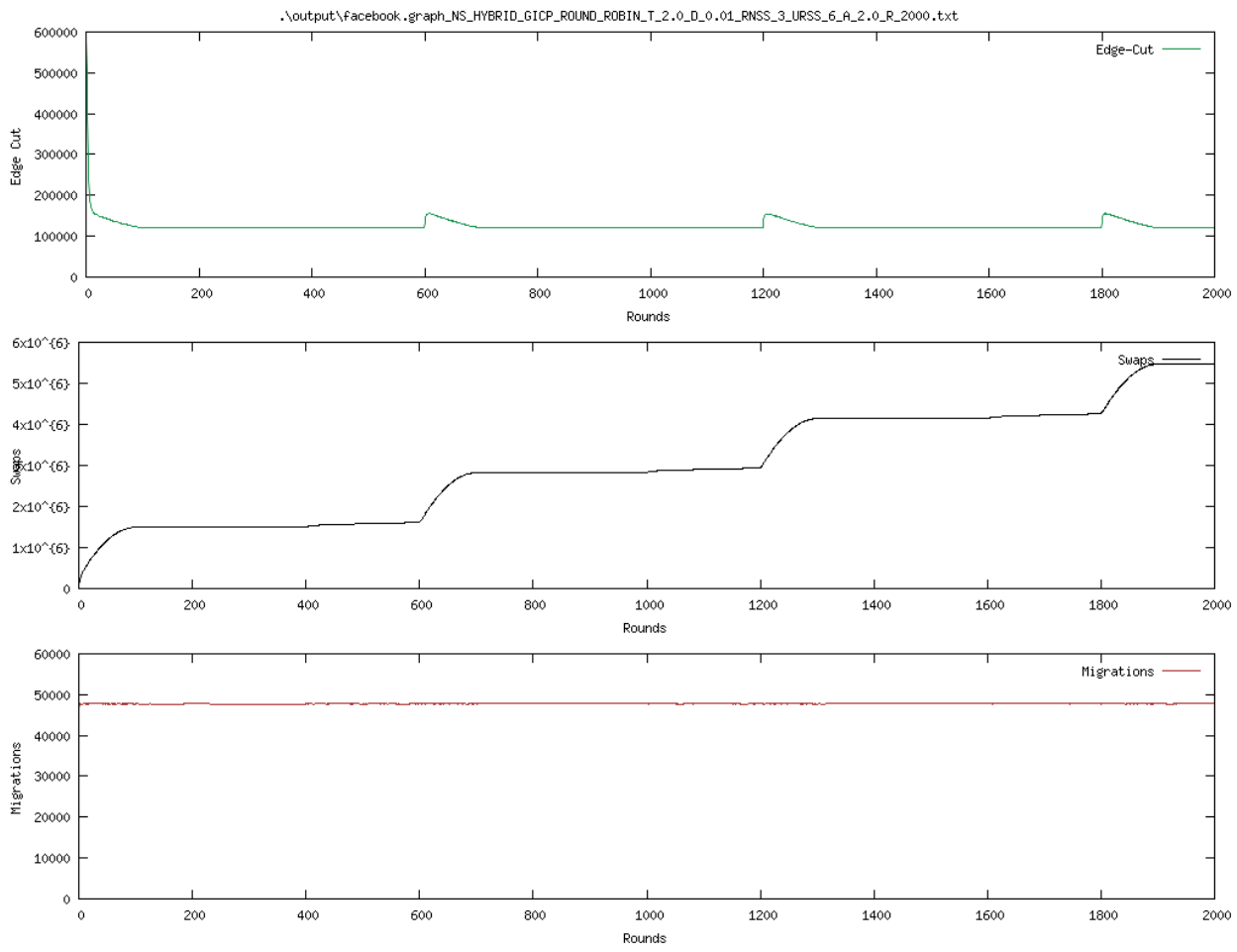
Edge-Cut
Edge Cut
Rounds
Swaps
Swaps
Rounds
Migrations
Migrations
Rounds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Edge-Cut
Edge Cut
Rounds
Swaps
Swaps
Rounds
Migrations
Migrations
Rounds